Circle field test

Print the markers

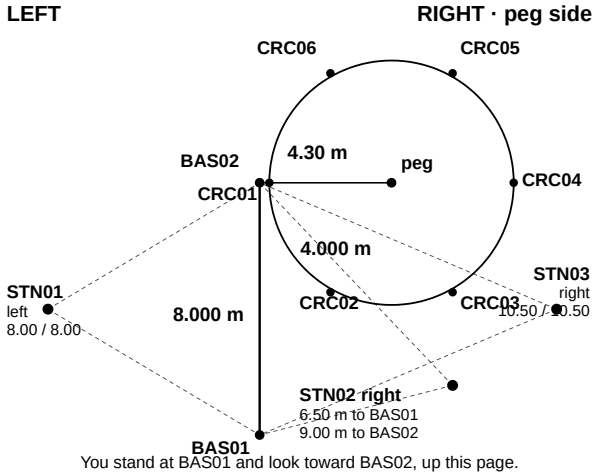
Print the sleeves the day before. Each sleeve is one A4 sheet. **Actual size / 100%**. No fit-to-page, no scaling, no borderless. Black only, highest quality. The **100 mm** bar must measure 100 mm (±0.5 mm), or reprint. Laminate first if you use film, then cut. Cut the left edge on **CUT HERE**, the right edge on the dashed line (or the sheet edge), and 5 mm off the top and the bottom. If the dashes or the band ink are clipped, cut at the edge of the ink. Do not crease the fold. Fasten the blank tab to the **60 mm** dowel first (flush staple or flush pin in wood, or glue or double-sided tape on a tube; no raised heads under the overlap), at least **4 mm** from the fold, never through the bands. Wrap snug, **TOP** up, and stick the cut left edge down over the tab. Use the **195 mm** fold crosses to keep the wrap square. Turn the seam away from the stations. Then R = C / 2π at three heights, to **0.1 mm**, on the cane row. Steps: How to print the markers.

All sleeves. BAS01 315, BAS02 36E, CRC01 459, CRC02 6BA, CRC03 8A3, CRC04 952, CRC05 DAC, CRC06 EC5.

STN01, STN02, and STN03 are where you stand. They are not sleeved. Calibration reuses a cane. This test does not use a checkerboard.

One person. One iPhone. About an hour. Numbers are the approved field sheet. Do not change them.

Look at the plan from **BAS01 toward BAS02** (up the page). **Left** is left. The peg is on your **right**.



You stand at BAS01 and look toward BAS02, up this page.

Kit

- ☐ One iPhone. Rear camera, zoom **1×**, held **landscape**. Same phone for every photo, including the calibration.
- ☐ A tape that reaches **8 m** clear. STN01 uses all of that 8 m.
- ☐ A laser, or a tape that reaches **10.5 m**.
- ☐ A non-stretch line marked at **4.000 m** against the tape.
- ☐ A centre peg. Six canes for the circle. Two canes for the baseline. One cane for the calibration if you do not reuse a circle cane.
- ☐ Sleeves from **Print the markers**, glued snug on a **60 mm** dowel. Matte. Each sleeve's own R.
- ☐ A small spirit level. The phone pole (about **1 m**). This sheet and a pen.

Camera location off. Leave the app's **Offset A (mm)** and **Offset B (mm)** at **0**.

Set out

1. Peg in the lawn. Line on the peg, **4.000 m** to the stick axis. Plant **CRC01** nearest the baseline.
2. Each next cane is **4.000 m** from the peg and **4.000 m** from the one before: **CRC02**, **CRC03**, **CRC04**, **CRC05**, **CRC06**. Closing chord CRC06–CRC01 within **30 mm** of 4.000 m, or walk them again.
3. Diameters CRC01–CRC04 and CRC02–CRC05, tape level, axis to axis **8.000 m**. Both within **20 mm**. Then write radius σ **3 mm**. Outside those limits: walk the sticks again. Do not write a larger σ.
4. From the peg through CRC01, **4.30 m** is **BAS02** (about **0.30 m** past CRC01). Write the distance you taped.
5. Baseline square to that line. From BAS02 back toward the peg mark **3.00 m**. From that mark swing **5.00 m**, and from BAS02 swing **4.00 m**. Through that side point, **BAS01** is **8.000 m** from BAS02.
6. Tape BAS01–BAS02 **both ways**, sleeve to sleeve, level. Level each reading, then add that end's measured R. The two axis lengths must agree within **10 mm**. The mean is the length you type. Do not average a pair that disagrees by more than 10 mm.
7. Laser the same baseline, sleeve to sleeve, same height. This is **CHK-BASE**. It is not typed into the app. Retake if the dot is more than half that sleeve's R (about **15 mm**) off the middle. Level, then add R for each sleeved end.
8. Bubble every cane (CRC01–CRC06, BAS01, BAS02, CAL). Middle of the bubble, about **1°**. A 2° lean at 0.5 m sleeve height is **17 mm**. Write height to the middle of the paper. **0.5 m** is a good target. Tick plumb only when the bubble passed.

Photos

1×, landscape, lens glass over the middle of the pole, pole bubbled, then shoot. Stay on the pole when you turn. A second photo is the same station. Do not step. If a listed mark is cut off, re-measure. Do not add a turn.

Calibration, before any station. One plumbed cane. Front of the phone about **4.000 m** from the paper (between **3 m** and **5 m**). Tape that, level, write the raw reading. Axis length is the reading plus that sleeve's R. One still. Call it **CAL** on this sheet.

STN01 — left. 8.00 m / 8.00 m.

1. Both baseline ends, and CRC01, CRC02, CRC03, CRC04, CRC05. CRC06 does not fit.
2. Turn. Keep **BAS02**. Add CRC06.

STN02 — right. 6.50 m from BAS01, 9.00 m from BAS02. Outside the circle.

1. Both baseline ends. No circle stick between them.
2. Turn. Keep **BAS02**. Take CRC02, CRC01, CRC06, CRC03. **Do not take CRC05.** It sits behind CRC03.
3. **Do not photograph CRC04 from here.**

STN03 — right. 10.50 m / 10.50 m. Required. Two stations are not a finished test.

1. Both baseline ends, and CRC01, CRC02, CRC03.
2. Turn. Keep **BAS02**. Add CRC06, CRC05, CRC04.

Tape each station from the **pole** to the **sleeve**, level. Add R once, for the sleeve end only. Write the readings below. The app has no box for them.

In the app

This build has no probe page. Shoot in **Camera.app**, then load with **Library**.

Before you leave the house, in Camera.app: **12 MP**, **HEIF Max** off, **48 MP** off, **ProRAW** off, location off.

1. Open Garden Planting. Tap the bottom-right menu button (**Open menu**). **Tools** → **New garden**, if this phone should not keep an older garden.
2. Menu → **Sign in** → **Sign in with Microsoft**. **Version** is the file name (**1** is **/Garden Survey/garden-v1.json**, shown on the path line).
3. Menu → **Baseline**. **End A BAS01**, **End B BAS02**. **Length (m)** is the tape mean after level and +R, not the laser. **Offset A (mm) 0**, **Offset B (mm) 0**. **Measure with Tape**. **Save baseline**. Do not swap the ends.
4. **+ Point**. The **Aiming preview** opens. Tap **Cancel**. In the **+ Point** dialogue: **Name Circle**, **Geometry Circle**, then **+ Item**. **Save**. There is no box for sleeve R, sleeve height, or plumb. They stay on this sheet.
5. For a station, **Cancel** the preview if it opens, set **Name** to **STN01** (then **STN02**, **STN03**, **CAL**), tap **+ Item**, then the **+** on the **Point** row (title **+ Point**).
6. On **Aiming preview**, do not use the shutter. **Take photo** is aim-only and has no lens data. Tap **Library**. Pick the Camera.app still at **Actual Size**.
7. Second still at the same station: shoot it in Camera.app without stepping, then **+ Photo** on that point, then **Library** again.
8. Tap the thumbnail. The screen is **Mark BAS01–BAS02**. Tap **A** on BAS01 and **B** on BAS02, then **Confirm**. **Target**, **Suggest**, **Clear**, and **Cancel** are the other buttons. There is no CRC button. Write which sticks were in the frame on this sheet.
9. The **Photo** row must read **verified**, and the line under it **Upload verified.** If it says **failed**, tap **Retry**. If the message is **Sign in to upload.**, menu → **Sign in** → **Sign in**. A line **Not usable for focal length.** means retake at 1× in Camera.app and pick **Library** at **Actual Size** again.
10. Menu → **Sign in** → **Save**. Wait until it says the file was saved to OneDrive. The path line is the file you send.

There is no left/right control. Tick the side on this sheet. Do not type 0 or 1. Do not type the laser, the check diameter, or the check chord.

Write on this sheet

Date _____ Phone _____ Lens name _____

	CRC01	CRC02	CRC03	CRC04	CRC05	CRC06	BAS01	BAS02	CAL
Height m									
Sleeve R mm									
Plumb	☐	☐	☐	☐	☐	☐	☐	☐	☐

Baseline BAS01→BAS02, level +R		Baseline BAS02→BAS01, level +R	
Pulls differ by (mm). Within 10 mm		Mean in Length (m). Tape, not laser	
CHK-BASE laser, sleeve to sleeve		Diameter CRC01–CRC04, sleeves	
Long chord CRC01–CRC03 (~6.93 m)		Peg to BAS02 (aim 4.30 m)	
CAL, phone front to paper		STN01 to BAS01 / BAS02	
STN02 to BAS01 / BAS02		STN03 to BAS01 / BAS02	

Side, looking from BAS01 toward BAS02: STN01 left ☐ STN02 right ☐ STN03 right ☐

Lens over the pole, then bubble: STN01 ☐ STN02 ☐ STN03 ☐

Which photo is CAL _____ Radius σ (3 mm if the diameters and the closing chord passed) _____

Send back

OneDrive path under **Version** in **Sign in**. Version **1** is **/Garden Survey/garden-v1.json**. Also **/Garden Survey/photos/** and **/Garden Survey/photos/manifest.json**, and a photo of this filled sheet. Every **Photo** row must say **verified**. Leave the stills unedited. Do not put them in git. If sign-in is not configured, **Tools** → **Export garden.json**, and still send the sheet photo.